Project Proposal: Submodular Optimization and Greedy Approximation for Distributed Influence Maximization

1 Topic Selection

We choose **Topic 3: Influence Maximization** from the course reference list. Influence maximization asks how to select a limited number of seed nodes in a social network so that information, behavior, products, or interventions spread to as many nodes as possible. It is a fundamental model for viral marketing, public-health interventions, and misinformation control, and it provides a clean algorithmic connection between modern networked applications and classical combinatorial optimization. This project focuses on **submodular optimization and greedy approximation for distributed influence maximization**. We study the problem from two complementary perspectives:

- **Algorithmic perspective:** How does the classical greedy algorithm exploit the submodularity of the influence function, and how can acceleration techniques such as lazy evaluation (CELF/CELF++) and reverse-influence sampling (TIM/IMM) reduce computation while preserving approximation guarantees?
- **Distributed perspective:** In large-scale networks, a single machine or agent may not have access to the entire graph. Each local agent observes a subgraph or estimates local diffusion effects, then communicates with other agents to approximate a globally effective seed set under a fixed budget. This setting naturally connects to distributed submodular maximization and multi-agent coordination problems.

2 Problem Formulation

Let $G = (V, E)$ be a directed or undirected graph with node set V and edge set E. Each edge (u, v) carries a propagation probability $p_{uv} \in [0, 1]$. Given a budget k, the goal is to choose a seed set $S \subseteq V$ with $|S| \leq k$ that maximizes the expected number of activated nodes:

$$\max_{S\subseteq V,\ |S|\leq k} \sigma(S),$$

where $\sigma(S)$ is the expected influence spread from S under a stochastic diffusion model.

Diffusion models. We consider two classical models:

- **Independent Cascade (IC):** When node u becomes active, it has a single chance to activate each inactive neighbor v with probability p_{uv}, independently.
- **Linear Threshold (LT):** Each node v draws a random threshold $\theta_v \in [0, 1]$; node v becomes active when the sum of incoming edge weights from its active neighbors exceeds θ_v.

Under both models, $\sigma(S)$ is monotone and submodular [1], meaning the marginal gain of adding a new node decreases as the seed set grows:

$$\sigma(A \cup \{v\}) - \sigma(A) \geq \sigma(B \cup \{v\}) - \sigma(B), \quad A \subseteq B,\ v \notin B.$$

Exact influence maximization is NP-hard. However, the classical greedy algorithm of Nemhauser et al. [2] achieves a $(1-1/e)$ approximation guarantee for maximizing any monotone submodular function under a cardinality constraint.

3 Related Work

Kempe, Kleinberg, and Tardos [1] introduced the influence maximization problem for social networks, proved NP-hardness under the IC and LT models, and showed that the greedy algorithm yields a $(1 - 1/e)$-approximate solution. This is our **core foundation** for connecting influence maximization with submodular greedy approximation.

Nemhauser, Wolsey, and Fisher [2] proved the classical approximation guarantee for maximizing a nonnegative monotone submodular function subject to a cardinality constraint. Their result provides the theoretical foundation for all greedy-based influence maximization algorithms.

Leskovec et al. [3] proposed the Cost-Effective Lazy Forward (CELF) optimization, which exploits the diminishing marginal gains of submodular functions to skip unnecessary evaluations. CELF achieves up to 700× speedup over naïve greedy while preserving the same approximation ratio. **Goyal, Lu, and Lakshmanan** [4] further improved this with CELF++, achieving an additional 35–55% speedup.

Tang, Xiao, and Shi [5] proposed TIM (Two-phase Influence Maximization), which uses Reverse Influence Sampling (RIS) to achieve near-optimal time complexity with a $(1-1/e-\varepsilon)$ approximation guarantee. **Tang, Shi, and Xiao** [6] subsequently proposed IMM (Influence Maximization via Martingales), which uses martingale concentration inequalities to further tighten the sample complexity and scale to networks with billions of edges.

Mirzasoleiman et al. [7] proposed GreeDI, a distributed protocol for submodular maximization under cardinality constraints. GreeDI partitions the ground set across multiple machines, runs local greedy on each partition, and merges the results. It achieves a constant-factor approximation and provides the main basis for our distributed optimization component.

4 Algorithm and Technical Plan

We plan to implement and systematically compare the following methods:

Baseline methods:

- **Random:** select k seed nodes uniformly at random.
- **Degree centrality:** select the k nodes with highest (out-)degree as a simple graph-structural heuristic.
- **PageRank-based:** select the k nodes with highest PageRank score.

Greedy methods:

- **Standard greedy:** at each step, select the node with the largest estimated marginal influence gain via Monte Carlo simulation.
- **CELF / CELF++ lazy greedy:** maintain upper bounds on marginal gains and skip re-evaluation of candidates whose upper bound is below the current best, reducing the number of costly influence estimations.

Scalable methods:

- **TIM / IMM:** generate random Reverse Reachable (RR) sets and select seeds by a weighted maximum coverage procedure. These methods achieve $(1 - 1/e - \varepsilon)$ approximation in near-linear time.

Distributed optimization component:

- **GreeDI-style distributed greedy:** partition the graph across m agents; each agent runs local greedy on its subgraph and returns its top-k candidates; a central coordinator merges the mk candidates and runs a final round of greedy selection. We will study how the number of partitions m and communication budget affect solution quality.

Datasets. We will evaluate on both synthetic and real-world networks:

- **Synthetic:** Erdős–Rényi random graphs, Barabási–Albert scale-free graphs, and Watts–Strogatz small-world graphs with varying sizes ($n \in \{500, 1000, 5000, 10000\}$).
- **Real-world:** NetHEPT (High-Energy Physics collaborations, ∼15K nodes, ∼59K edges) and Wiki-Vote (Wikipedia adminship votes, ∼7K nodes, ∼104K edges).

Evaluation metrics: expected influence spread (estimated via Monte Carlo), wall-clock runtime, number of influence evaluations, and scalability with respect to graph size n, seed budget k, and number of Monte Carlo samples R.

5 Project Scope and Expected Goals

Core reproduction goals:

- Greedy selection achieves significantly larger influence spread than random, degree, and PageRank baselines, validating the effectiveness of submodular optimization.
- CELF/CELF++ preserves nearly the same solution quality as standard greedy while reducing runtime by one to two orders of magnitude.
- TIM/IMM achieves comparable influence spread with further runtime improvements, demonstrating the advantage of reverse-influence sampling.
- Influence spread increases with seed budget k, while marginal gains decrease monotonically, confirming the submodularity of σ.

Distributed optimization goals:

- The GreeDI-style distributed scheme achieves a reasonable approximation ratio compared to centralized greedy.
- Solution quality degrades gracefully as the number of partitions m increases, illustrating the tradeoff between communication cost and approximation quality in distributed submodular optimization.

Optional further extensions:

- Compare the IC and LT diffusion models on the same graphs.
- Analyze the effect of Monte Carlo sample size on estimation variance.
- Implement a communication-limited consensus protocol where agents exchange partial information over multiple rounds.
- Discuss connections to distributed optimization with local information and global objectives.

6 Team Information

Member	Responsibility	Expected Contribution
Yixin Chen	Modeling and theory	Problem formulation, related work, and complexity analysis
Bichi Zhang	Algorithms	Greedy, CELF/CELF++, TIM/IMM, GreeDI implementation and analysis
Yaoqin Ye	Experiments	Datasets, evaluation, plots, and distributed optimization analysis

References

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[3] Jure Leskovec, Andreas Krause, Carlos Guestrin, Christos Faloutsos, Jeanne VanBriesen, and Natalie Glance. Cost-effective outbreak detection in networks. In *Proceedings of the 13th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 420–429, 2007. https://doi.org/10.1145/1281192.1281239.

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